UNIT I - FPGA Based Systems

Introduction to FPGA-Based Systems

FPGA (Field Programmable Gate Array) is a programmable semiconductor device based on configurable logic blocks (CLBs) connected via programmable interconnects. Features: Reconfigurability, parallel processing, high speed, low development cost. Applications: DSP, Embedded systems, Communication systems. Advantages: Short time-to-market, flexibility. Limitations: Higher power consumption compared to ASIC.

```
+----------------------+
|      I/O Blocks      |
+----------------------+
|  Configurable Logic  |
|       Blocks (CLB)   |
+----------------------+
| Programmable Routing |
+----------------------+
```

FPGA Architectures

Architecture consists of LUTs, Flip-flops, switch matrices, clock management units, BRAM and DSP slices. SRAM-based FPGAs are volatile and widely used. Anti-fuse FPGAs are one-time programmable. Flash-based FPGAs are non-volatile.

```
+-----+
| LUT |
+--+--+
   |
+--v--+
|  FF |
+-----+
```